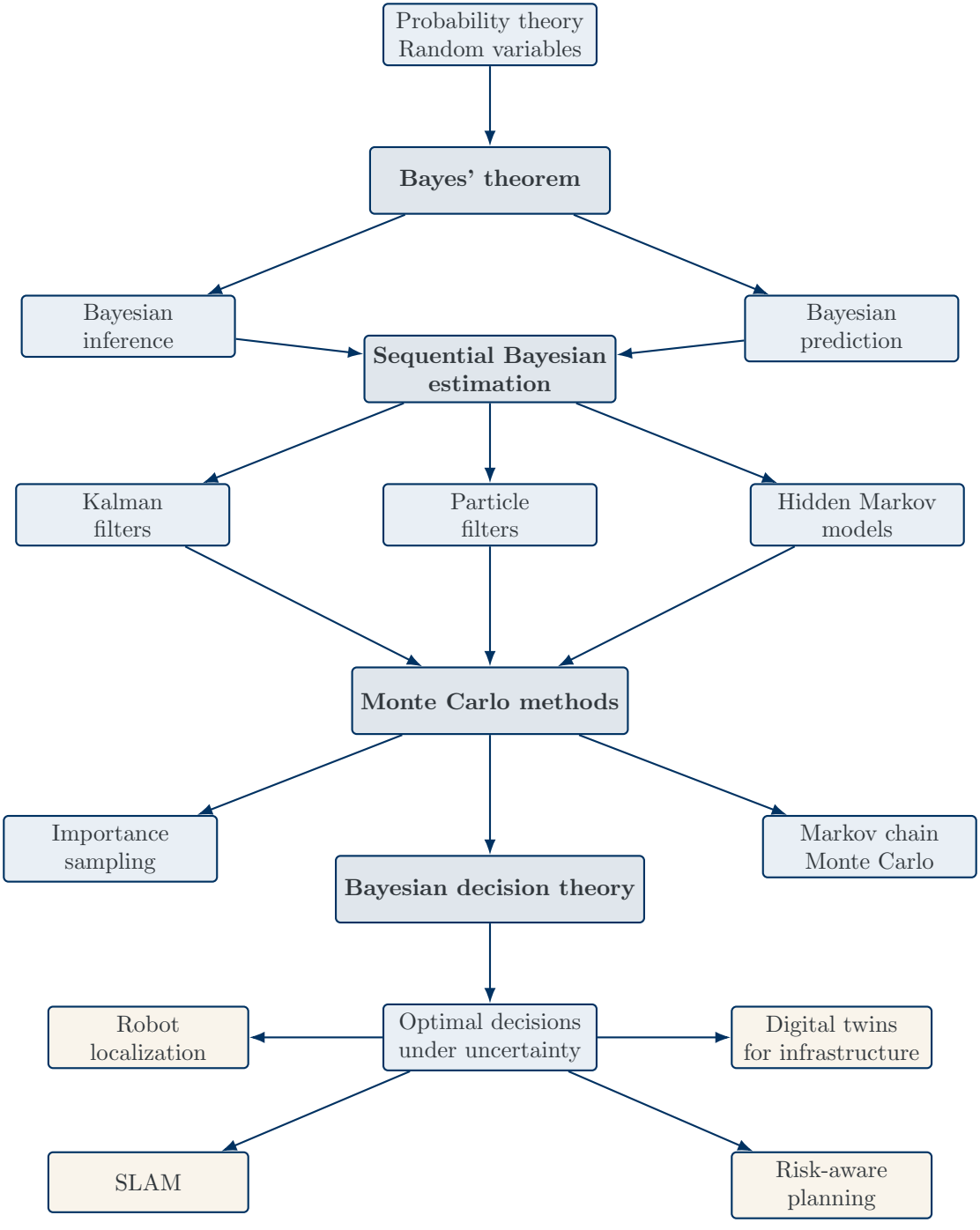
Probability theory
Random variables
Bayes' theorem
Bayesian inference
Bayesian prediction
Sequential Bayesian estimation
Kalman filters
Particle filters
Hidden Markov models
Monte Carlo methods
Importance sampling
Markov chain Monte Carlo
Bayesian decision theory
Robot localization
Optimal decisions under uncertainty
Digital twins for infrastructure
SLAM
Risk-aware planning